

Today's access technologies

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The drive to deliver an increasingly diverse range of telecommunications products over an access network, designed originally for the delivery of telephony, has demanded, in recent years, the development and deployment of a range of copper, fibre and radio-based delivery solutions. This paper outlines the history and the current status of some of the key access delivery solutions that exist within BT's portfolio of access components today.

1. Introduction

The drive to reduce the operating cost of BT's access network, both in terms of the ongoing capital investment and the manpower required to operate it, is ever present. This requirement, when set against a backdrop of the increasing customer expectations for improved functionality and reliability, has triggered the development and deployment of a number of delivery solutions within the access domain over the past few years. Focusing the finite available development resource within access on to the key developments demands a critical analysis of the strategic fit as well as the costs and benefits associated with any proposed new developments at the feasibility stage. This approach ensures that the number and variety of new access solutions being introduced into the access network to provide the desired feature set is always kept to an economic minimum.

While emerging copper-enhanced technologies promise the delivery of higher and higher bandwidths over the existing copper access network, fibre and radio-based delivery solutions still figure prominently in the access component strategy. This paper gives a view of the access technologies in use today, which, although they may eventually be superseded by emerging technologies discussed in other papers within this edition, will still form a major part of the network for some time to come.

2. Narrowband delivery solutions

Narrowband is the general term used to cover 64 kbit/s services but $n \times 64$ kbit/s up to 1984 kbit/s have, in general, also been included. This section describes some of the key technologies for delivering these narrowband services.

2.1 Digital Access Carrier System (DACS)

From shared service to DACS

For almost as long as there has been a telephone network there has been demand for service that could not be

satisfied due to the lack of copper pairs in the access network. Over the years various methods have been devised to circumvent this bottle-neck. The simplest method was 'shared service' or 'party line' which allowed two customers to share a line, but allowed only one call to be present at a time.

During the early 1970s a 1+1 carrier system (WB900) was developed. BT still uses this system today but it is now obsolescent and the number of systems in use is declining rapidly. A digital 1+1 system (WB1900) was developed during the late 1980s to supersede WB900 but was abandoned in favour of a fully digital system. Other approaches include local line concentrators and street-sited multiplexers.

Development of DACS began in the early 1990s. The first variant is designated DACS1 and does not support caller display. This was superseded by the current version, DACS2. Deployment in significant volume commenced in 1995.

Description of DACS

DACS may be described as a 'back-to-back' access delivery system. At the local exchange two analogue PSTN lines are cabled from the MDF to the DACS exchange shelf. In the exchange unit two lines are multiplexed into 2B+D format and cabled back to the MDF (see Fig 1).

At the DP the line is terminated on a DACS remote unit (RU). The process is reversed and the two analogue PSTN lines are cabled to the customer's premises.

Exchange equipment

There are two suppliers of DACS equipment to BT but distinct differences exist in the physical realisation of the two suppliers' exchange equipment which prevent successful interworking. The exchange shelf provided by

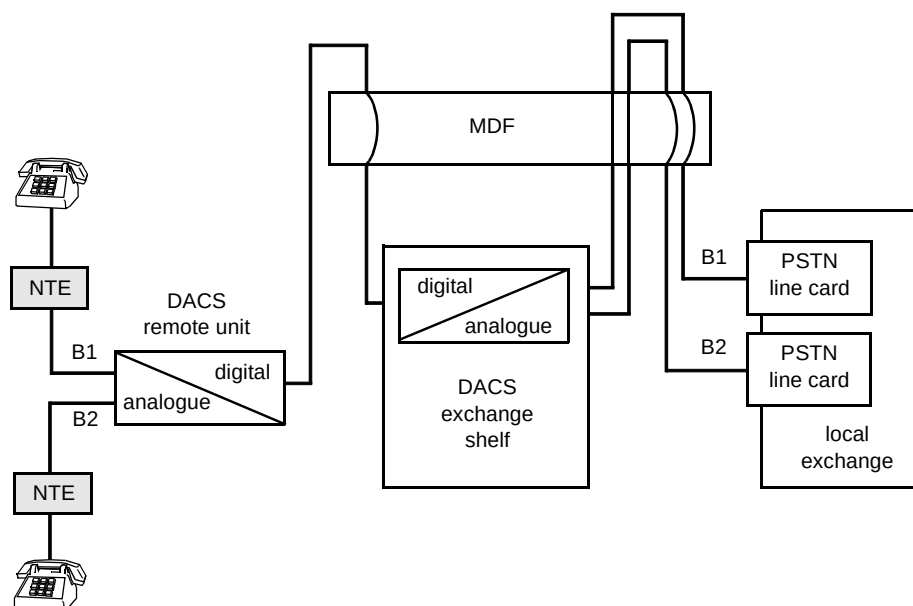


Fig 1 The digital access carrier system.

one supplier contains a common card and fifteen exchange unit (EU) cards. Each EU terminates four PSTN interfaces, and two digital line interfaces. This gives a total of 60 circuits and requires 30 access network copper pairs.

The exchange shelf provided by the second supplier contains a common card, eight analogue line cards (ALCs) and eight digital line cards (DLCs). (ALC/DLC card pairs are also generically referred to as an EU.) Each ALC terminates ten PSTN interfaces and each DLC terminates five digital line interfaces. This gives a total of 80 circuits and requires 40 access network copper pairs.

Within the EU is a bypass circuit to enable the exchange equipment to be installed on a working 'donor' PSTN circuit with minimal disruption, e.g. during jumpering. The EU then remains dormant until it detects the presence of an RU. When an RU is detected, the EU becomes operational, the bypass relays operate, and the pair gain system begins to operate.

Remote equipment

There are three types of remote unit — pole/wall mounting, footway box mounting, and internal. The electronics for each is the same, but the design and materials used for the external casing are optimised for the environmental conditions prevalent for each application. For safety reasons not more than four DACS RUs may be mounted on a pole.

Line testing

Because of the digital transmission system, it is not possible for a conventional PSTN line test system to have visibility of the line beyond the analogue port on the EU. To provide an indication of the status of the line between the

RU and the NTE/CPE, a mimic system is employed. The RU has a simple test tool that enables it to determine the terminating conditions on its analogue ports, e.g. NTE and CPE (working circuit), or low resistance A to B (insulation fault). This information is relayed over the D channel to the EU. When stimulated by a PSTN test head the EU presents a mimic impedance best representing the RU line condition to the test head. To speed up the testing process the RU automatically tests the line every 70 minutes and the result is stored at the EU.

Powering

Power for the remote unit is fed from the exchange unit at 140 V dc, with the current limited to 40 mA.

Planning limits

The maximum recommended EU/RU insertion loss is 37 dB @ 40 kHz. This is equivalent to 4.1 km of 0.4 mm cable or 6.4 km of 0.5 mm cable (5.9 km). The maximum loop resistance of the cable between the EU/RU is 1250 Ω . The maximum loop resistance of the 'drop' between the RU and the customer premises equipment is 200 Ω .

Services supported

DACS2 provides the necessary signalling to support a direct exchange line (DEL) including select services and caller display. It does not provide the extra signalling required for private branch exchanges (PBXs). Additionally, because there is not a continuous DC path from the LE to the customer, services such as meter pulse facility, high impedance loop detection (30 k loop), and Telecom Red alarms cannot be supported. DACS1 does not support caller display.

Modems

BT does not offer any guarantee of the data rates that may be achieved across the PSTN. The level of performance is dependent upon the quality of the transmission path, e.g. low noise and the compatibility of the terminal equipment. Due to the additional A-to-D and D-to-A conversions performed in the DACS equipment, modem performance may be reduced when compared to a good-quality direct copper connection. Typically data rates will be reduced by 2 kbit/s for lines routed over DACS. Table 1 gives details of the number of DACS deployed.

Table 1 Volume of DACS deployed (March 1998).

System size	No of DACS
No of working lines served by DACS	493 000
No of DPs with DACS remote units	211 000 (5.3% of DPs)

Why use DACS?

In a competitive trading environment BT cannot be complacent about the way it responds to requests for provision of service. Neither is it practical nor economically viable to build a copper network that will always have capacity exactly where it is required. When an order has been taken and it is not possible to 'route first time from stock', there are a number of options available, e.g. divert pairs, use 'stopped pairs', reactive network build, repair known faulty pairs or deploy DACS. DACS is used where it is the most cost-effective solution or where it is not possible to meet the agreed delivery date by other means.

Evolution

Demand for pair gain systems will continue for the foreseeable future. In recent months a number of feasibility studies have been conducted to determine how such systems might evolve. There is synergy between DACS and HomeHighway and there may be scope to redevelop the HomeHighway NTE9 into a new generation RU. This would significantly reduce the complexity of the exchange

equipment. Systems with a higher gain, e.g. 0 + 4, are appearing on the world market, although the use of these may be constrained by the limitations of line powering the RU.

2.2 HomeHighway

The BT lowband digital access (HomeHighway) provides ISDN basic access services and analogue PSTN service through the same copper access network connection to the customer.

HomeHighway is primarily intended as a conversion product to existing customers, but also has the ability to be supplied as a new provision. The service facilitates an upgrade to customers' PSTN installations by enhancing the functionality of the existing line to provide an additional digital capability of 2×64 kbit/s data.

The HomeHighway development is a collaborative venture between BT and the major local switch manufacturers (Fig 2) which replicates the existing PSTN service, additionally offering a high-speed digital path, and enabling a customer to:

- have a maximum of two simultaneous calls (digital or analogue, incoming or outgoing),
- retain PSTN service, directory number, select services, discounts, home wiring and CPE,
- benefit from the switched digital connectivity at data rates of 64 kbit/s or 128 kbit/s (128 kbit/s is achieved via applications running on CPE),
- access ISDN supplementary services on the data channels.

In general terms, the HomeHighway service is enabled by the provision of two newly developed items of equipment, one of which is provided by core network and the other as an addition to the customer premises access network termination. The core network utilises the recently

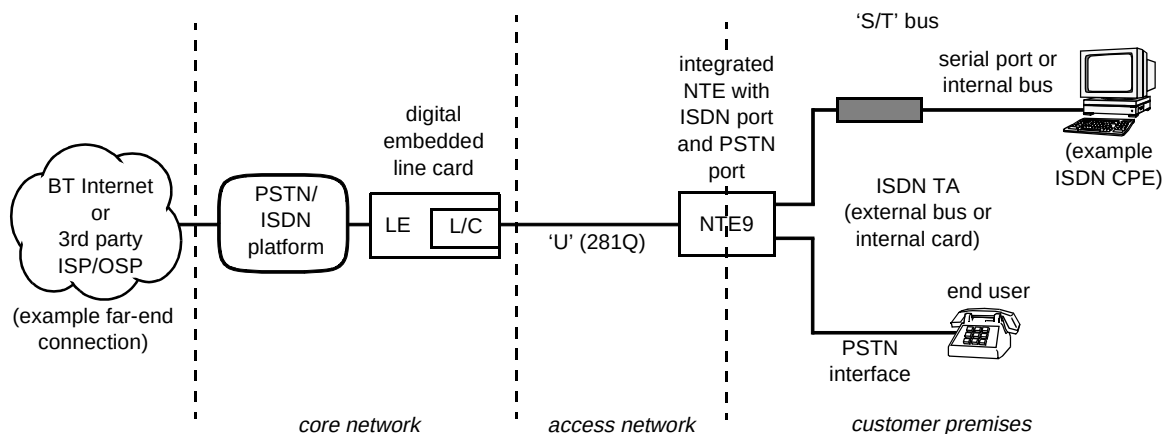


Fig 2 HomeHighway.

introduced embedded digital line card while the access network requires the provision of an additional network termination equipment — NTE9 (Fig 3).

The NTE9 is connected to the existing NTE5 master socket replacing the face plate. The NTE9 provides the electronics which enable the detection of analogue signals to be mapped (using a convention that originates from the ETSI V5.1 standard) to appropriate information elements within the digital signalling frame. The core network Home-Highway enhancements enable the appropriate call control mechanisms (i.e. PSTN call control platform, ISDN call control platform) of the digital local exchange to respond to the received signalling information elements of the digital signalling message protocol.

Main PSTN service features

The directory number (DN) for the PSTN service will be different from the DN of the ISDN service. The PSTN interface will remain with similar characteristics to a normal direct exchange line. In particular, the PSTN interface will:

- interwork with the existing PSTN CPE,
- provide adequate ringing current to support REN of 4,
- support CLASS signalling (i.e. caller display, etc),
- support both MF and loop disconnect signalling.

Currently the PSTN interface will not:

- support earth calling terminal equipment,
- support SPM, p-wire earth or 30 k Ω loop signalling,
- support RedCARE or RedABC alarm services.

Main ISDN service features

The HomeHighway service provides ISDN2 service at the ETSI standard.

2.3 The Digital Private Circuit Network (DPCN)

DPCN was established in 1983 for the provision of digital private circuits (KiloStream). KiloStream services were provided covering 2.4, 4.8, 9.6, 48 and 64 kbit/s data rates. From these early days the network has evolved to provide additional new services, i.e. KiloStream *N* and analogue-over-digital (AoD), as well as the deployment of remotely managed access multiplexers/switches and their dependent local access systems.

From 1983 to 1992 the Equipment Digital Muldex 6003 (EDM6003) was the only access multiplexer used in the network but from early 1992 the Equipment Network Access 1A (ENA) began to be deployed. The EDM6003 is a non-managed 31-channel primary multiplexer. Local customer access was provided over copper pairs from a tributary card, in the EDM6003, via a modem unit, to a customer-located network terminating unit (NTU). The modem unit was housed in a separate equipment (Equipment Modem (EM) TG2619) mounted alongside the EDM6003. In the early days the local line system used a 4-wire WAL2 line code but in 1986 2-wire WAL2 systems began to be deployed.

The customer-located NTU was locally powered either from the 240 V AC mains either as a stand-alone desk-top unit or shelf mounted. The EDM6003 2 Mbit/s aggregate port was connected to a manual crossconnect site. This consisted of a number of back-to-back EDM6003s interconnected via a distribution frame at 64 kbit/s (co-directional). It was thus possible to interconnect any one 64 kbit/s channel

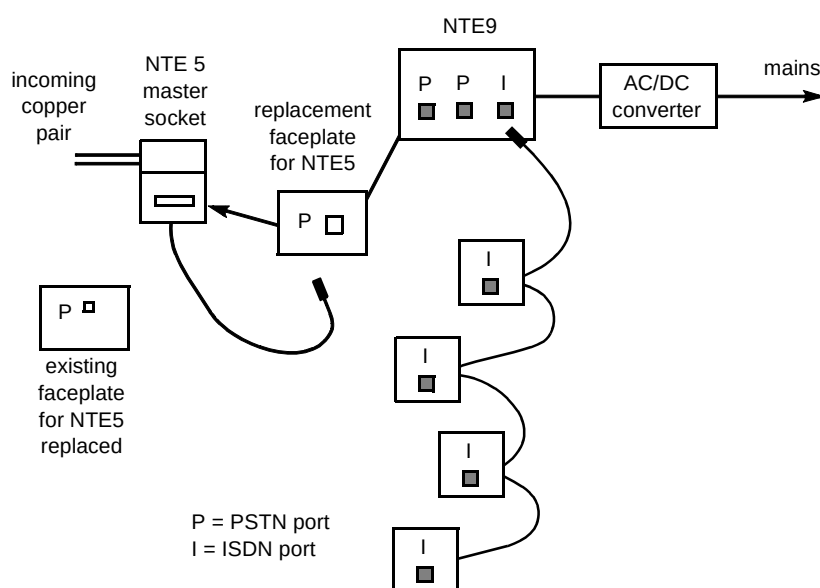


Fig 3 The use of NTE9 in the access network.

to any other items of across the country. Various other items of equipment provided the synchronisation and alarm-gathering facilities.

Both the EDM6003 and EM/TG2619 used TEP-1E equipment practice, each with a height of 487.68 mm. Also during 1985 a new remotely managed Automatic Crossconnect Equipment (ACE) began to be deployed, this was to eventually replace all the earlier manual cross-connect sites.

During 1986 a KiloStream *N* service was launched using a new EDM6003 trib card, 2 Mbit/s local line system and a customer located NTU (NTU1A later to be called the NTE1A). Although the equipment could support values of *N* from 2 to 30 only the values of *N* from 2 to 16 were sold as part of the KiloStream *N* service (except for international circuits where *N* up to 30 is deployed).

Use of Equipment Network Access 1A (ENA).

The deployment of the ENA, from 1992, has allowed a fully managed DPCN to be evolved. The ENA is managed by the Remote ENA and ACE Control Equipment (RENACE) via the 2 Mbit/s link between the ENA and the ACE (i.e. using a spare bit (8) within the TS0 Not Frame) (Fig 4).

Use of extension shelves, with the main ENA shelf allows for 91×64 kbit/s circuit-end equivalents to be configured. The ENA and extension shelves use type 88 equipment practice, each of height 396.24 mm. Thus this provides a considerable space saving over the earlier

EDM6003 and EM/TG2619 combination. New ENA line cards and NTUs have been provided, covering the earlier range of KiloStream and KiloStream *N* services plus some additional ones, i.e. KiloStream 2.4, 4.8, 9.6, 19.2, 48 and 64 kbit/s, KiloStream *N* Data, X.21 interface, KiloStream *N* voice fractional 2 Mbit/s G.703/704, analogue over digital (AoD), 2 and 4-wire analogue.

The KiloStream and AoD services are provided via the Line Card 2A (2-channel unit). Customer access is via the NTU6 for KiloStream and NTU11 for AoD. Desktop and shelf-mounted variants are provided. The provision of the ENA allowed local turn-around circuits to be provided as well as those for through connection to the ACE. Test facilities were built into the ENA in the form of a Test Card 3A. This allows testing of any circuits that terminate or pass through that particular ENA. A new Line Card 1A (single channel) was also provided to allow use of the NTU6 and NTU11 with the earlier EDM6003 and EM/TG2619 combination. The local line system for KiloStream/AoD uses an AMI line code running at 71.1 kbit/s.

Customer-located ENA

Where large quantities of private circuits are required to be delivered to a customer's premises it is possible to provide ENAs and extension shelves at the customer's site. This delivery method is called 'MultiStream'.

As well as using the normal range of line cards and NTUs, two new 'embedded' NTUs have been developed. These plug directly into the shelves and provide access for

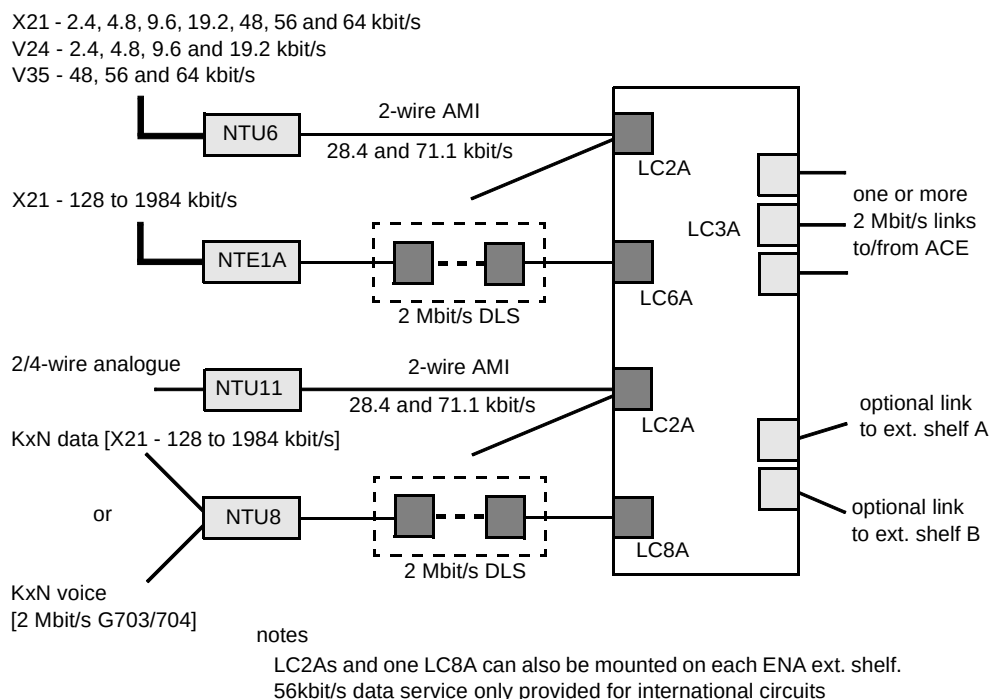


Fig 4 ENA 1A facilities.

KiloStream (NTU10A) and AoD (NTU9A) services. Each providing two channels per card.

KiloStream *N* provision on ENA

Early deployment of KiloStream *N* (data) utilised a Line Card 6A, a 2 Mbit/s digital line system and the earlier NTE1A. In 1994 the Line Card 8A (LC8A) and NTU8 were deployed allowing provision of KiloStream *N* data and *N* voice services. A 2 Mbit/s digital line system (DLS) was used to provide the local access. The NTU8 comes in desktop and shelf-mounted variants. All current provision of KiloStream *N*, via the ENA, utilises the LC8A/NTU8. Only five LC8As can be used on the ENA main shelf plus one in each extension shelf.

KiloStream Aggregate Service

This is provided where customers require multiple circuit provision via a 2 Mbit/s pipe. This utilises a direct connection from the crossconnect switch via a 2 Mbit/s DLS direct to the customer. The customer's interface conforms to ITU-T Recommendations G.703/704. Only KiloStream 64k and KiloStream *N* data (with no X.21 C/I signalling) services are currently supported.

Reduced Cost KiloStream *N*

The use of a 2 Mbit/s local line system for KiloStream *N* service provision, and its attendant cost, has been seen to be a barrier for customers wishing to migrate from 64 kbit/s to 128 kbit/s or higher.

In late 1998 BT will be deploying a new ENA Line Card 7A (LC7A) and its associated NTU7. Local access will be over a single copper pair, thus achieving considerable savings through the avoided deployment of a dedicated 2 Mbit/s DLS (Fig 5).

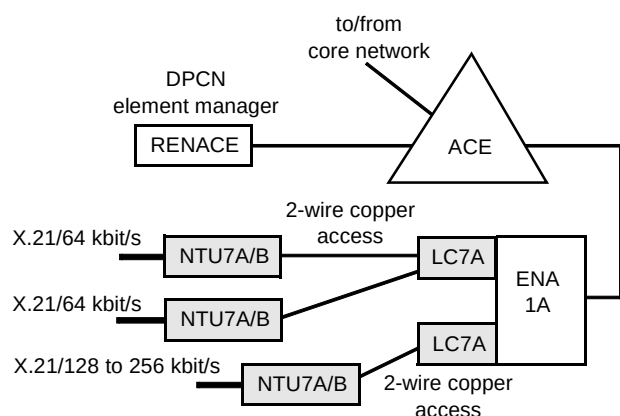


Fig 5 Reduced Cost KiloStream *N* provision.

This equipment will be used primarily to provide KiloStream *N* data services (X.21) at rates of 128, 192 and 256 kbit/s. This data range currently represents a considerable portion of BT's KiloStream *N* provision. Through transmission of the customer's C/I signalling information, it is a configurable option as per the LC8A/

NTU8. The LC7A provides line powering for the NTU7. The transmission reach is expected to be around 4 to 4.5 km (for the $n \times 64$ kbit/s service) which is only slightly less than that provided for the KiloStream service. For $n \times 64$ kbit/s use the LC7A provides access for one circuit end. In addition to KiloStream *N* data provision the LC7A/NTU7 can provide access for KiloStream 64 kbit/s (X.21) services. In this case two channels per LC7A are provided. For 64 kbit/s working it is possible to provide extended transmission reach (typically 6 km). Line powering of the NTU7 is also provided, but, at the full range (5–6 km), a local power supply unit, at the NTU7, is required. The line system uses a 2B1Q line code operating at 134 kBaud for $n \times 64$ kbit/s, and 34 kBaud for 64 kbit/s working.

The NTU7 comes in desktop/wall mounted (7A) and shelf-mounted versions (7B). It has a reduced range of facilities when compared with the NTU8, but has been engineered to provide a cost-effective unit with the same or better quality of service targets. It is also easier to operate. The NTU7A is less than two thirds the size of the existing NTU8A, i.e. 166 mm (w) $\times$ 173 mm (d) $\times$ 45 mm (h) (NTU8A = 251 mm (w) $\times$ 274 mm (d) $\times$ 63 mm (h)) (see Fig 6).

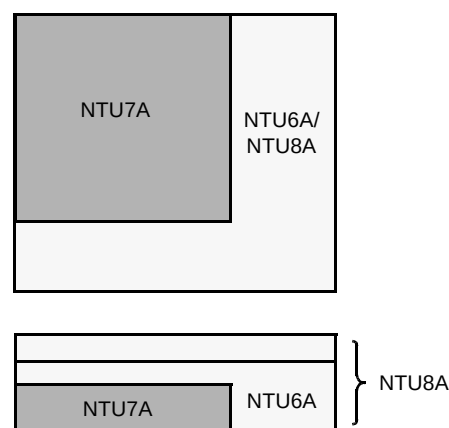


Fig 6 Comparison of NTU sizes.

LC7As can be configured in any ENA position that is used by the LC8A, but can also be used within each of the ENA extension shelves (7/8 per shelf), thus allowing greater ENA capacity for KiloStream *N* use.

Provision of LC7A/NTU7 for 64kbit/s use will also allow for easier migration to KiloStream *N* (2 to 4) because the customer's NTU will not have to be changed.

Summary

The DPCN has evolved over the past 15 years from a non-managed to, predominantly, a remotely managed network. The types of service supported have evolved from data only (64 kbit/s and below) to the additional support of analogue and KiloStream *N*.

Nationwide coverage has been achieved with a local exchange presence now at approximately 6000 sites. Since

1992 over 12 000 ENAs have been deployed. Crossconnect switches are currently located at approximately 100 sites. The DPCN is also now used to transport other services or to provide the local access connection. These include the Flexible Bandwidth Service (FBS) and the Switched Multi-megabit Data Service (SMDS). KiloStream and AoD services are also supported on TPON and on the London Local Fibre Network (LLFN). These facilities are fully compatible with that provided over the DPCN.

What about the future? The provision of sub-64 kbit/s services is now on the decline, while KiloStream *N* data services are currently experiencing substantial growth. In the longer term there will be an increasing use of other technologies and service platforms, e.g. ATM based, for the provision of private circuits.

2.4 DECT

BT are currently in discussion with OfTel and DTI with a view to using digital enhanced cordless telecommunications (DECT) for delivery of service to residential customer premises using radio systems. The proposed DECT system uses the same channels as that used by CPE DECT, i.e. the Diverse 1010. Studies have now been completed to show that public and private systems can co-exist in the same locality without affecting the capacity of either system, provided cell sizes are limited. The target areas for a DECT serving system are high-density, high-rise and multi-occupancy residential blocks. Studies show that, in these areas, fault rates can be three times higher than the national average. In many cases these areas are also identified as two-man-party areas where BT has decided, for staff safety, two people should attend faults. It may also be used where the demand for the 'In Contact' has caused a requirement for additional copper infrastructure. The DECT system will normally provide basic telephone services although low-speed modem and group 3 fax connections are possible. A typical DECT installation is shown in Fig 7.

The base-station may be mounted on a pole or wall with its service area restricted to 200 metres by obstruction from the served building or by antenna radiation patterns. The signal will penetrate the outer fabric of the building but signal losses will not normally allow penetration through the building. Power is fed to the base-station from the base-station controller which may be located in street furniture. Connection to the exchange may be any form of 2 Mbit/s line system. Where the served area is close to the exchange, the base-station controller may be located in the exchange. BT's licence does not allow provision of a mobile service and therefore the customer equipment must act as an NTE to allow the customer to connect his own equipment, which may be another cordless unit.

3. Wideband delivery solutions

Wideband is the term used to cover 2 Mbit/s, and above, delivery solutions. This section describes some of the key technologies used by BT to provide these.

3.1 MegaStream

MegaStream digital private circuits provide permanent, full-time connections between locations throughout the UK to carry high volumes of data and speech combined. MegaStream is available at speeds of 2, 8, 34, 45, 140 and 155 Mbit/s. Typically, it is used at 2 Mbit/s and 8 Mbit/s as the backbone of large corporate networks (Fig 8).

MegaStream 2, a 2 Mbit/s service, was launched in the early 1980s using standard first-generation junction PCM 30-channel equipment with minor variations related to power feeding and special housings for customers' premises. The difficulties associated with the transmission of 2 Mbit/s on existing local cables meant that specially developed digital transmission cables of transverse-screen design had to be installed. The first-generation systems were soon superseded by second-generation systems which used a 1.5 Mbit/s (4B3T) line rate for improved transmission performance. These systems were further

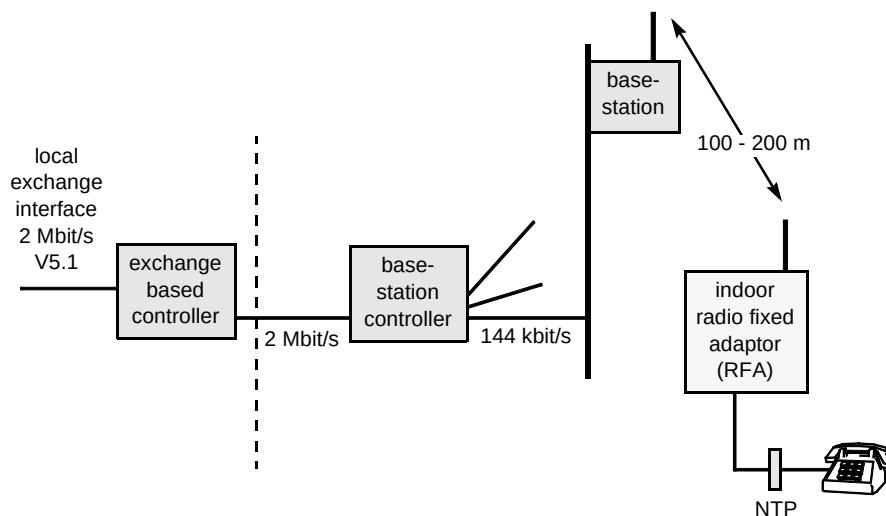


Fig 7 DECT solution proposed for PSTN access to high-rise, multi-occupancy buildings.

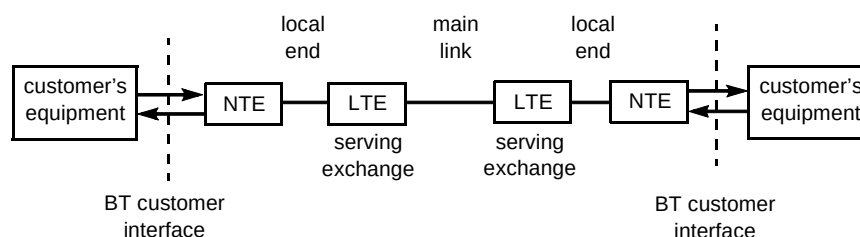


Fig 8 Typical MegaStream circuit.

adapted for MegaStream delivery by providing alarm reporting and fault-diagnostic facilities.

Although transverse-screen cable was a readily available cost-effective method of providing 2 Mbit/s service, this medium was not easily adapted for use above 2 Mbit/s. The increasing demand for private services at rates greater than 2 Mbit/s has resulted in optical fibre being the main method of delivering these services. There are also some applications where, irrespective of bit rate, the special attributes of optical fibre, for example, electrical isolation, make fibre the inevitable choice. In addition, the cost reductions on optical fibre systems in the mid-to-late 1980s also resulted in fibre eventually being used for direct 2 Mbit/s applications.

Until the mid-1990s the optical-fibre and copper applications in the access network made use of cable and equipment designed principally for the junction applications. During 1995 BT introduced 1 and 4×2 Mbit/s digital wideband serving section (DWSS) equipment, designed specifically for the access network. At the time, the main driver was the need to introduce EMC-compliant equipment to deliver BT digital wideband services prior to the enforcement of the EEC directives on 1 January 1996.

3.2 Point-to-point optical fibre — DWSS

DWSS is a transmission product that provides 2 Mbit/s MegaStream and ISDN 30 services economically over point-to-point optical fibre. DWSS is very reliable, and simple to install, commission and maintain. There are a number of suppliers of DWSS providing various capacities of 1, 4 or 16×2 Mbit/s over a fibre infrastructure. These systems have a high level of integration thus resulting in a low number of configuration modules and hence significantly reduced inventory cost. The basic architecture of DWSS, from the different suppliers, is very similar. DWSS is made up of the following modules (see Fig 9):

- exchange shelf, housing a number of line terminating equipment (LTE) — one per system and a common transmission network surveillance (TNS) card per shelf,
- network terminating equipment (NTE) located at the customer's premises.

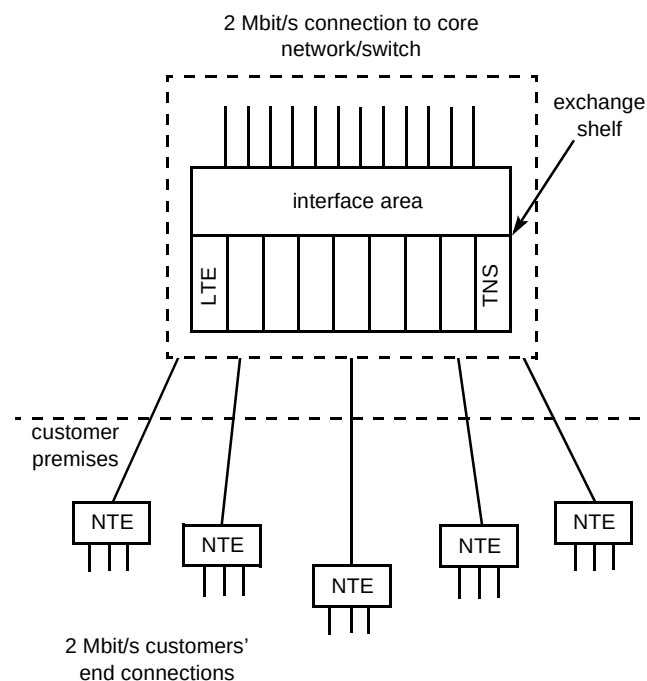


Fig 9 DWSS architecture.

The shelf is designed for TEP 1E or ETSI racks. All the 2 Mbit/s interfaces to the shelf are connected via an interface area using BT type 43 co-axial connectors.

There are currently two suppliers of 4×2 Mbit/s systems but only one supplier of the 1×2 Mbit/s and 16×2 Mbit/s systems. In 1997 an enhanced version of the 4×2 Mbit/s product was launched by one supplier which reduced the unit costs, allowed for the support of new services, i.e. X.21 data interfaces, and offered a more aesthetically pleasing unit at the customer's premises. The second supplier of 4×2 Mbit/s is planning to launch their new version in December 1998. Similarly, in order to meet the same objectives on 16×2 Mbit/s systems as achieved on the 4×2 Mbit/s, a new 16×2 Mbit/s product was launched in April 1998. This product is supplied by an existing 4×2 Mbit/s manufacturer and uses a common exchange shelf for 1, 2, and 16×2 Mbit/s systems.

3.3 Point-to-point radio

MegaStream and other customer wideband service deliveries are extensively provided by 18 GHz point-to-point radio systems with a capacity of either 4×2 Mbit/s or

16 × 2 Mbit/s. These radio systems require a line-of-sight path between a BT network site and the customer premises. Although it is normal practice to provide only a single radio link for a customer access link, in certain special cases two radio links can be provided in tandem, via a third party site. Link lengths of up to about 20 km are achievable using the 18 GHz systems with dish sizes of up to 600 mm diameter.

Radio is generally used to provide the customer access link under the following circumstances:

- a temporary expedient to enable provision of service within the agreed contractual or required customer service time-scales when a permanent cable-based solution cannot be provided in time,
- a permanent solution where the cost of providing service via cable exceeds the cost of provision by radio,
- to provide diversity or separation of service as required and agreed with the customer — radio avoids the need to use common duct and even a common BT serving site in certain cases in order to achieve maximum physical separation of the alternative path.

The use of radio wideband access links has been particularly effective in recent years in supplying MegaStream circuits to the more remote base-station sites of mobile telephone operators.

3.4 SDH

BT delivers MegaStream private circuits across a range of technologies, including plesiochronous digital hierarchy (PDH) and synchronous digital hierarchy (SDH). Due to the increasing importance of SDH-based services to customers, BT has launched a range of SDH MegaStream services at 2, 8, 34, 45, 140 and 155 Mbit/s. BT's SDH platform has been designed to offer MegaStream products which are highly resilient. Network configuration and capacity can be provided rapidly, allowing urgent customer bandwidth requirements or anticipated surges in network traffic to be changed within a matter of days without any interruption to existing services.

MegaStream Genus is BT's high-availability MegaStream service delivered over resilient SDH systems using optical fibre. MegaStream Genus is used for critical applications where speed and security are paramount. Customers' sites are connected via two physically separate paths with all cables carried in separate ducting. Should a fault develop, network traffic will be immediately redirected along the alternative path.

3.5 Point-to-point copper (CWSS)

In 1997 BT introduced copper wideband serving section (CWSS) for the delivery of 2 Mbit/s services over copper pairs. CWSS allows BT to capitalise on the inherent capability of its existing metallic access network. CWSS uses high-rate digital subscriber line (HDSL) technology to provide the same functionality as the highly successful DWSS equipment, but uses metallic pairs as the transmission medium. CWSS supersedes the HDSL stage-two technology which has been used since 1995.

CWSS provides the following advantages:

- shorter planning and installation time-scales compared with the previous version of HDSL equipment,
- substantial cost savings over DWSS (typically 80%), achieved principally through avoided fibre infrastructure build,
- improved provision processes, simpler installation procedures and the availability of spare copper pairs in the local network has enabled the number of 2 Mbit/s circuits delivered per month to double,
- operational and management capabilities brought into line with DWSS delivered systems.

Due to the reduced costs of delivering ISDN 30 over CWSS, the minimum number of channels that can be economically delivered can be reduced from 15 to 6 or even lower. This has allowed the price of fractional ISDN primary rate services to fall within reach of a greater number of small-to-medium-sized businesses.

Currently, BT provides between 10% and 30% of its new 2 Mbit/s provisions over CWSS.

CWSS systems operating over a two-pair system can be used for most situations. Three-pair systems are used where increased transmission range is required.

The main limitations of CWSS compared with DWSS are substantially shorter range and reduced robustness of the transmission medium. DWSS will operate over a distance of up to 30 km whereas CWSS has a maximum range of about 4 km over transverse-screen cable, reducing to under 3 km on standard 0.4 mm local cable. The metallic network is less reliable than fibre delivery for the following reasons:

- existing installation and repair working practices for metallic pairs are based on analogue services and not specifically 2 Mbit/s services,
- more frequent intervention by working parties,
- metallic cables are more prone to faults arising from environmental conditions.

3.6 Products and services

The introduction of the next generation of radio and SDH systems to complement the recently introduced DWSS and CWSS systems will provide BT with a complete portfolio of wideband multiservice delivery systems within the access network. Products and services capable of being supplied are ISDN 30, KiloStream *N*, MegaStream, CellStream (BT's new ATM-based service) [1], DealerStream, Featurenet, SMDS (Switched Multi-megabit Data Services) [2], IP (Internet protocol), Frame Relay and FBS (Flexible Bandwidth Service).

4. Telecommunications over a passive optical network — TPON

In simple terms the access network provides a connection between a customer's equipment and the telecommunications equipment in the local exchange. The traditional view is that of a wholly copper cable environment from 'telephone to exchange'.

Customers' premises house a network termination point which normally comprises a modern socket into which CPE, such as a telephone, would connect. The exchange houses the transmission terminating equipment.

In between these two points the general case is that of copper wires in multiwire cables carried in underground ducts providing a physical and electrical means to connect the two points together. The wires carry electrical impulses which provide both signalling for call control and the transmission bandwidth of the service required from the network by the customer.

Physical and electrical limits are imposed by the copper telecommunications media. Electrical limits are often expressed in basic 'Ohms law' terms, higher resistance having a profound effect upon the voltages and current required to secure adequate signal and transmission quality.

Physically these cables require not only a multiplicity of connection points throughout the access network but also accessibility for maintenance purposes as the fault liability of a traditional access network using copper is still too high. It has long been an aim to simplify the access network, but in doing this there are two main drivers to consider:

- to make telecommunications services more attractive to the customer,
- to create an access network which is cheaper, more reliable and easier to manage for the network operator.

An effective way to make customers more attracted to the services obtainable from the network is to provide a connection that can offer a variety of network services not

limited by bandwidth or physical constraints such as distance from the serving exchange. In an attempt to meet these customer requirements while also satisfying the second main driver by introducing a rigorous operations administration and maintenance functionality, BT developed TPON.

4.1 TPON access network technology

The main achievements realised through the introduction of TPON are:

- the introduction of fibre into the access network,
- element management of the access bearer technology,
- access network configuration changes achieved without the need for extra line plant.

Fibre optics offers a high capacity medium where, instead of using electrical signals, the transfer along the fibre is achieved in the form of pulses of light. The chosen wavelength 'window' where wavelengths of light propagate most efficiently with minimum attenuation through single mode optical fibre is 1300 nm. Despite the inherent very high capacity of fibre optics, it is still relatively expensive compared with copper access networks as a result of the geographically distributed nature of the bandwidth requirement across the access network. The advent of passive optical networks ('passive' because no opto-electrical conversion occurs at any point within the distribution other than the termination points) has, however, meant that, under certain deployment situations, competitive gains over the traditional copper approach can be made.

A PON consists of a single fibre pair originating from an optical line termination (OLT) at the exchange. At points along the fibre it 'splits' to produce more fibres. This splitting process repeats and effectively duplicates the topology of traditional access networks (Fig 10). However, unlike the traditional access networks, the fibres are terminated by an optical network unit which can take advantage of the higher bandwidth of fibre and deliver the possibility of a multiplicity of telecommunications services on one bearer (a single fibre pair).

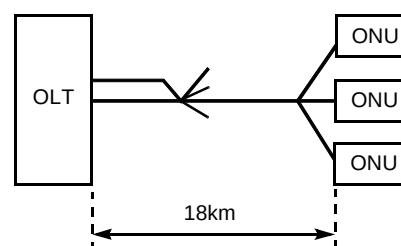


Fig 10 The PON structure.

4.2 Architecture

A typical TPON deployment (Fig 11) comprises an OLT (optical line termination at the exchange) and several ONUs (optical network unit at the customer's premises) with the following modularities:

- each OLT can support:
 - up to 32×2 Mbit/s interfaces with core network,
 - up to 4 PONs (with each PON being able to support 256×64 kbit/s equivalent services),
 - up to 24 ONUs per PON,
- while each ONU can support:
 - up to 32×64 kbit/s equivalent services,
 - up to 2×2 Mbit/s G.703-based services.

The notable maintenance tool which separates TPON functionally from the traditional copper access network is the TEM (TPON element manager). The TEM has been designed and developed with the capabilities to:

- manage all aspects of the ONU/OLT,
- record equipment build levels,
- manage circuit provision,
- conduct testing,
- perform alarm management,
- enable faster service provision/migration,
- remove the need for MDF/PCP jumpers.

Using the existing network operational tools in concert with the TEM (Fig 12), the network operator can respond virtually 'on-demand' with the ability to:

- change a customer's services
- alter the delivered service capacity (bandwidth).

4.3 TPON-installed base

There are now more than 6000 lines of TPON in service providing mainly PSTN service. Both analogue and digital private circuits can also be provided, as can ISDN2.

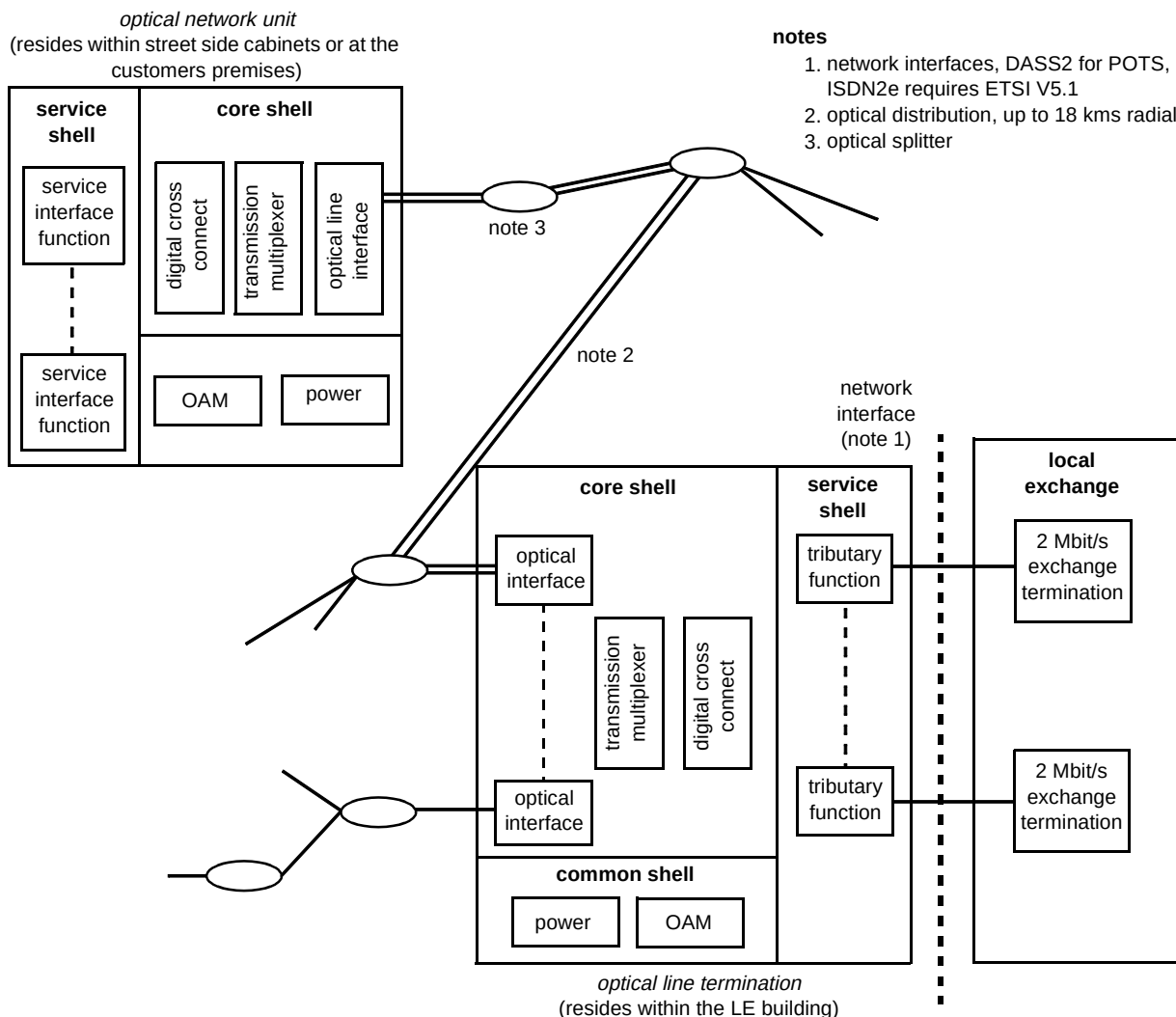


Fig 11 A typical TPON deployment.

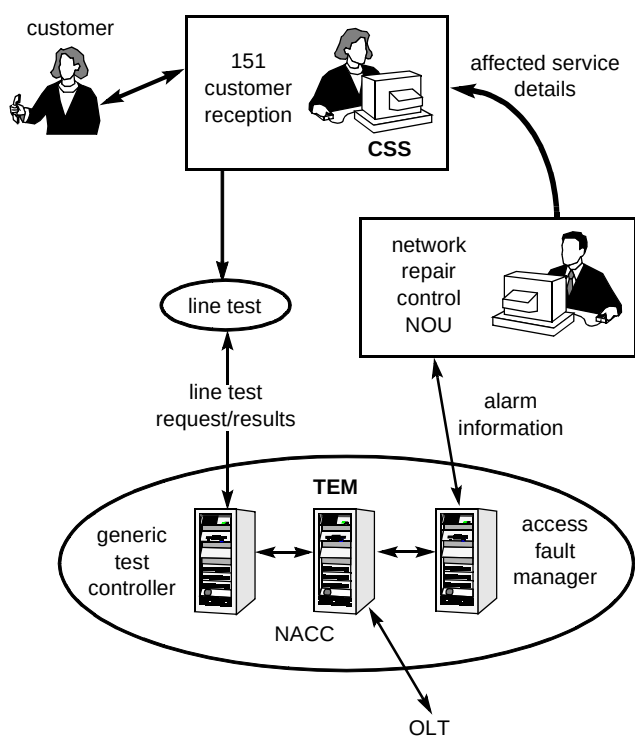


Fig 12 Use of TEM/network operator to control TPON functionality.

For the circuit-switched services of the PSTN and ISDN, TPON is dependent upon the network connection to the relevant service platform. DASS2 (network) is used in this respect while ETSI V5.1 will be used in future for provision of ISDN2e.

4.4 ETSI V5.1 interface

The ability of TPON to provide a multiplicity of telecommunications services from the network is derived from the 2 Mbit/s network connection. While DASS2 provides an ample package of analogue PSTN services by a proprietary BT signalling mapping, the same cannot be said for the latest ISDN requirements. With the launch of ISDN2e (ETSI service level) TPON has adopted the new 2 Mbit/s connection provided by the European standard ETSI V5.1 interface to retain its ability to support the core network service platforms.

Primarily designed by ETSI for access networks utilising PON technology, the V5.1 interface is a static multiplexer using a common ISDN signalling frame envelope to support multiservices, in particular analogue PSTN and digital ISDN.

The ISDN2e (all ETSI-compliant) services of:

- basic call (introduced by ETSI ISDN),
- CLIP and CLIR,
- DDI,
- MSN,
- MCID,

- sub-addressing,
- terminal portability,
- videotelephony,
- 7 kHz telephony,

will all be able to be provided via TPON when the network introduces these services.

4.5 Next steps for TPON

The introduction of the ETSI V5.1 interface will not lead in the short or medium term to the abandonment of the DASS2 interface. The PSTN service level provided by this interface can offer a suitably large variety of analogue services without modification. The notable exception will be the introduction of enhanced analogue services such as 'telemetry'. The ETSI V5.1 interface can also be developed to provide this service variant in time.

The next major opportunity for the further development and deployment of TPON will be the launch of BT's new HomeHighway service. The ETSI V5.1 interface will be the delivery vehicle for TPON and the initial co-operation between TPON and the HomeHighway engineering development has assured a compatibility across the access network technology aspects of the expected service.

5. Conclusions

This paper has provided an overview of some of the key technologies being used by BT in the access network and each solution has been developed to provide a cost-effective response to a particular set of demands. As the portfolio of products grows, there is inevitably some overlap which offers a choice of solution in an increasing number of cases. The decision as to which solution to use depends on a number of factors and this information is embodied in a set of planning and other service-specific implementation rules which are determined during the design phase of the system. The strict use of this information is imperative in getting the right solution in place to satisfy the requirements of both BT and the customer. The factors which influence the 'rules' are many and varied and it is clear that, as technology evolves and more solutions become available, choice of the correct solution will be vital.

The access network accounts for a large proportion of BT's capital investment with millions of individual connections to customers. The variation in customer types, the large numbers involved, and the huge investment already made, mean that the access network is particularly sensitive to the solutions that are adopted. The need to satisfy not only today's demand but also future demand is a careful balancing act between cost and flexibility. However, the competition and other financial pressures dictate the need to drive the cost of access network provision downward, while at the same time demanding improvements to

the quality of service and the number of services carried. Thus costs will continue to have a major influence on which technology solution is chosen/used. As investment in the access network grows, this becomes an ever more difficult equation to balance and solution designers must take this into account when deciding how best to satisfy a particular demand.

Acknowledgments

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Fred Adams joined BT in 1968 as a technician in the Research Department at Dollis Hill. He initially worked on a number of transmission and digital design projects. On promotion in 1979 he moved to work on the development and deployment of 30-channel PCM systems and later had responsibility for the development and launch of the new KiloStream services. He now works within the access strategy and design unit (part of *networkcapacity*) where he heads up a team responsible for narrowband and wideband access solution design. For a number of years now he has also had

particular design responsibility for all narrowband private circuit access equipment.



Geoff Gauntlett joined in 1971 as a technician in the transmission development department. He initially worked on cable television making, installing and commissioning headend equipment. In 1976 he took the opportunity to transfer to a team building a prototype access multiplexer. This was followed by a string of projects involving the innovative use of electronics in the access network. In 1986 he joined the team developing the London local fibre network. Since promotion in 1991 he has been involved in the design and specification of PSTN access equipment. He now works

within the access strategy and design unit (part of *networkcapacity*) where he has design responsibility for narrowband PSTN access equipment.



Duncan Marland joined BT as an apprentice in Preston telephone area in 1963. After receiving a BTech degree in Electrical and Electronic Engineering from Bradford University in 1969, he began working initially on television transmission systems. In 1973 he moved to working on digital data private networks.

On promotion in 1982 he continued to work on various projects associated with the development of KiloStream services.

He now works in the access strategy and design unit (part of *networkcapacity*) where he has design responsibility for wideband private circuit access equipment. He is a member of the IEE and a Chartered Engineer.



John Baker joined BT in 1966 as an apprentice. He has worked in the AC signalling group where he was involved with miniaturisation design and development of signalling systems.

Later he joined the local signalling system strategy department and was instrumental in early ISDN access signalling protocol design and development (DASS1, DASS2, DPNSS, I-Series, etc) and represented BT at European standards forums.

He is now part of the access strategy and design unit of *networkcapacity*. He is a solution designer and has responsibility for the delivery of TPON technology. A major portion of his work is now concerned with the introduction of the ETSI V5.1 interface on the local switch and within the access network.



Don Wenham joined in 1963 as an apprentice and worked in London on the maintenance of transmission systems.

On promotion, he became the radio maintenance officer responsible for radio maintenance in London. During the same period he controlled the team responsible for the installation and maintenance of radio structures and antennas.

He transferred in 1986 to take on the responsibility for the policy and performance of radio in the access network. He is an incorporated engineer and a member of the IEE